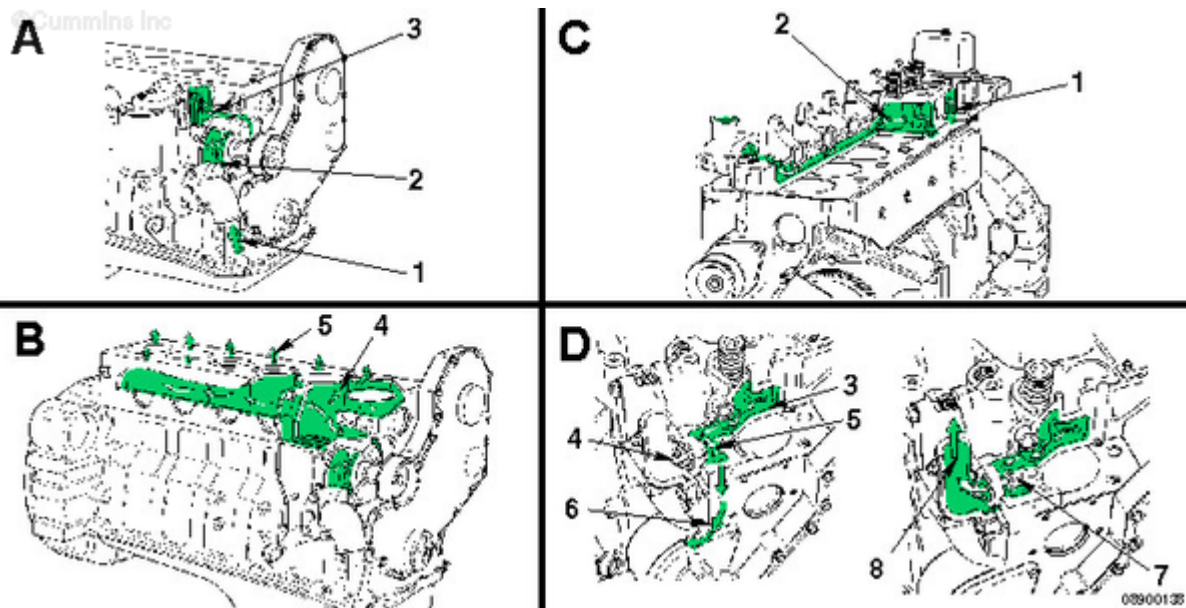


Trainings ▾

General Information



⚠ CAUTION ⚠

Never operate the engine without a thermostat. Without a thermostat, the coolant will not flow to the radiator and the engine will overheat.

The illustration identifies the significant features of the coolant system.

Sections A and B.

Coolant is drawn from the radiator by the integrally-mounted water pump. The output from the water pump empties into the oil cooler cavity of the cylinder block.

The coolant then circulates around each cylinder and crosses the block to the fuel pump side of the engine.

1. Coolant Inlet
2. Pump Impeller
3. Coolant Flow Past Oil Cooler
4. Coolant Flow Past Cylinder Head
5. Coolant to Cylinder Head

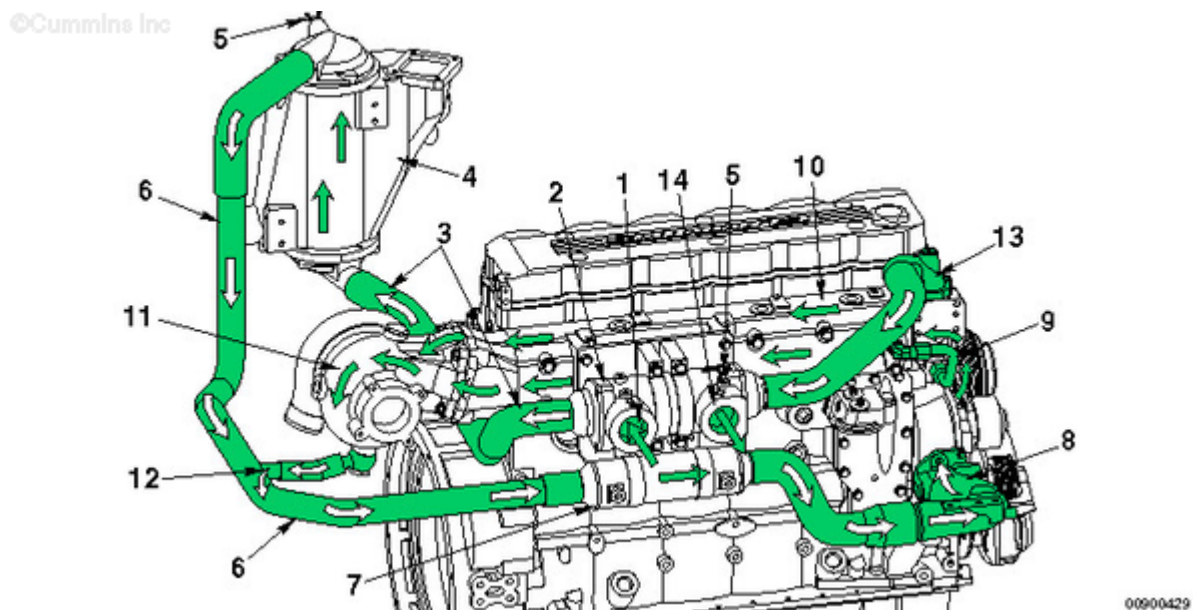
Sections C and D.

Coolant then flows up into the cylinder head, crosses over the valve bridges, and down the exhaust manifold side of the engine to the integral thermostat housing.

As the coolant flows across the head toward the thermostat housing, it provides cooling for the injector. When the engine is below operating temperature, the thermostat is closed, and the coolant flow bypasses the radiator and goes to the water pump inlet through internal drillings in the block and cylinder head.

1. Coolant Flow from the Cylinder Head
2. Coolant to the Thermostat Housing
3. Coolant Flow Past Injector
4. Thermostat
5. Coolant Bypass Passage
6. Coolant Flow to Pump Inlet
7. Bypass Closed
8. Coolant Flow Back to Radiator

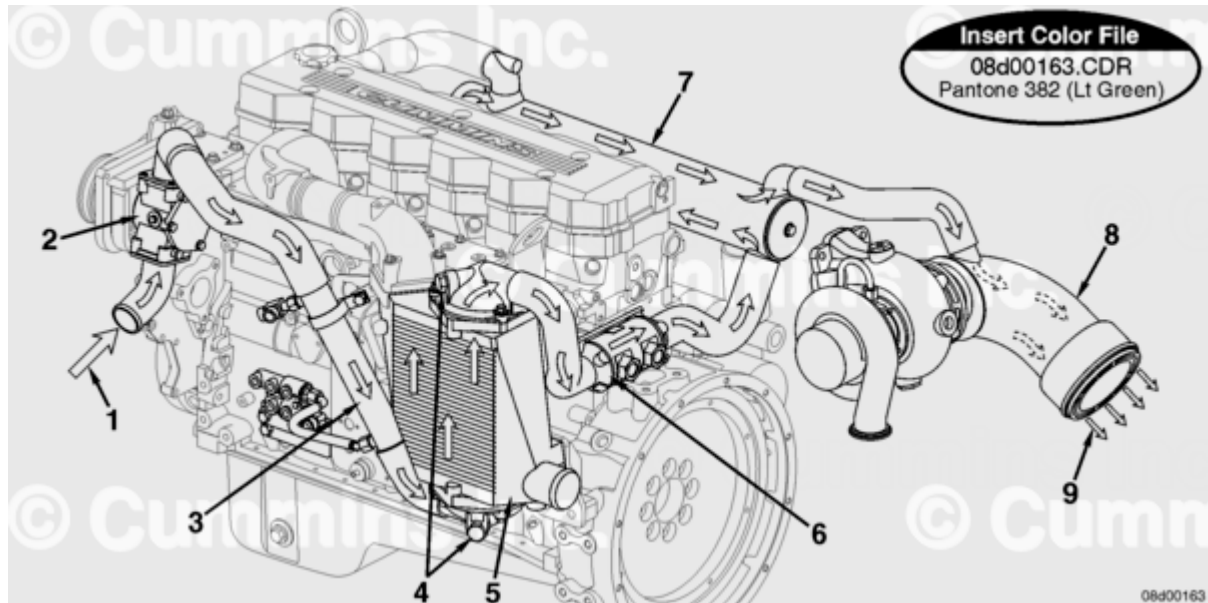
When operating temperature is reached, the thermostat opens, blocking the bypass passage to the water pump and opening the outlet to the radiator.



Engine Coolant from, Keel Cooled (Exhaust Side)

1. Coolant inlet from keel cooler
2. Thermostat housing for keel cooling
3. Coolant inlet from thermostat housing to aftercooler
4. Aftercooler
5. Engine cooling system fill vent petcock
6. Coolant outlet from aftercooler to marine gear cooler
7. Marine gear cooler
8. Coolant inlet to water pump

9. Engine coolant outlet to exhaust manifold
10. Water cooled exhaust manifold
11. Turbocharger with water cooled turbine housing
12. Turbocharger coolant outlet
13. Engine coolant outlet to thermostat housing
14. Coolant outlet to keel cooler



Sea Water Coolant FLOW

1. Sea water supply
2. Sea water pump
3. Fuel cooler
4. Aftercooler zinc anode
5. Aftercooler housing
6. Marine gear oil cooler
7. Heat exchanger
8. Exhaust outlet
9. Sea water discharge